

TAKUYA BOEHRINGER

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Full-stack robotics engineer who takes systems from CAD to deployment on real hardware: mechanisms, electronics, perception, controls and robot learning. Builds to accuracy and runtime specs and validates on physical robots. British and Canadian.

WORK EXPERIENCE

- Robotics Engineering Intern - Animo Robotics - (Perception)** May - July 2026
- Built the **perception stack** for a robotic cell, using a stereo depth camera to create a voxelized point cloud of different foods in robot-frame, with **PCL**, **Open3D** and **OpenCV**.
 - Developed the **calibration procedure** and an extensive **testing harness** to validate depth accuracy under various conditions, including different cameras.
 - Achieved targeted accuracy of $\pm 2.5\text{cm}$ in x,y, $\pm 1.5\text{cm}$ in z and run time under 100ms.
 - Ran an accuracy study to compare cameras, and made the camera an interchangeable component for future improvements.
 - Leveraged **SAM** to localize the conveyor belt running past the robot in real time.

- Summer Researcher - UCL - (Robot Perception and Learning Lab)** May - Sept 2024
Collaborated with Prof Dimitrios Kanoulas: paper accepted by the IEEE International Conference on Automation Science and Engineering (CASE).
- Presented the paper live at **CASE 2025** in Los Angeles.
 - Developed a **Human-in-the-Loop** system, sending the cartesian coordinates of the VR controllers over **ROS2** to the controller on the robot, sending back calculated joint angles.
 - Built a VR interface in **Unity** which integrates **Gaussian Splatting** and takes in the calculated joint angles, using them to copy the manipulator position on a URDF in the environment.
 - Validated the user interface via a 15 person **user study**.

- Software Engineering Intern - Leonardo - (Electronic Warfare)** July - Aug 2023
- Contributed to a classified project in the **Electronic Warfare** division.
 - Programmed in C++ for **embedded linux programming** and to interface with firmware.
 - Designed a **WPF user interface** using **socket programming** to interact with embedded devices.

PROJECTS

- Columbia Intelligent Construction Lab (ICON)** Sept 2026 - Present
- Currently building a drone-manipulator system to autonomously inspect screws with an ultrasonic contact probe.
 - Designing the manipulator and attachment to the drone frame in **Onshape**.
 - Designing and tuning an **INDI controller** for the drone.
 - Transferring the simulation from **Gazebo** to **Isaac Sim** to train a **reinforcement learning** policy, leveraging **domain randomization**.
- Columbia Robotic Manipulation and Mobility Lab (ROAM)** Feb - May 2026
- Contributed to the **MiniBEE** project, a bimanual robotic manipulator using two reduced-mobility arms coupled into a kinematic chain.
 - Developed automated kinematic optimization across joint configurations under **Prof Matei Ciocarlie**.
- Autonomous Robotic-Arm Docking System for Spacecraft** Jan 2026 - May 2026
- Designed an RPR (revolute-prismatic-revolute) robotic manipulator for the Intro to Human Spaceflight class taught by former astronaut **Prof Mike Massimino**, and built a small-scale prototype.
 - Converted the CAD model from **Onshape** to a URDF in **Isaac Sim**, training a **reinforcement learning** policy on it to grab an object.
- Robot Studio Project** Sept - Dec 2025
- Engineered and **3D printed** a legged robot for the Robotic Studio class by **Prof Hod Lipson**.
 - Utilized **Fusion 360** to design the whole assembly, including two **Klann linkages** for eight-legged locomotion and electronics housings.
 - **Soldered** all the electronics including the buck converter and the motor controller board.
 - Trained it to crab walk in **MuJoCo** using **reinforcement learning**, attempting **REINFORCE** but finding that **PPO** performed better in simulation.
 - Deployed the policy **sim-to-real**, achieving a real-world walking speed of 15.79 cm/s.
- Electrochemical Machining Drill** Sept 2025 - Nov 2025
- Designed and built a 1 dimensional ECM machine, using electrolysis to drill into aluminum workpieces.
 - Designed the stepper motor circuit on **KiCAD**, moving the tool at a calculated material removal rate.

- Implemented safety protocols including E-stop, fusing strategy and power supply bench validation.

PUBLICATIONS

[Conference Paper] **IEEE International Conference on Automation Science and Engineering** *First Author*
Immersive Teleoperation Framework for Locomanipulation Tasks

[Journal Paper] **Scientific Reports (Nature)** *Second Author*
Analyzing Spatio-temporal Dynamics of Dissolved Oxygen for the River Thames using Superstatistical Methods and Machine Learning

EDUCATION

MS Mechanical Engineering (Robotics and Control Track) - Columbia University August 2025 - Present

MEng Computer Science (Robotics Minor) - University College London (UCL) Sep 2021 - June 2025

SKILLS

Integration & Deployment: *ROS2, embedded Linux, Docker, Git, hardware interfaces (sockets, firmware), sim-to-real deployment, ONNX, test harness design*

Programming: *Python, C++, C, C#, MATLAB*

Perception: *PCL, Open3D, OpenCV, ICP, RANSAC, point clouds, depth cameras, camera calibration, segmentation (SAM)*

Controls & Learning: *PID and INDI control, Reinforcement Learning (PPO, REINFORCE), domain randomization, PyTorch, supervised and unsupervised learning*

Simulation: *MuJoCo, Isaac Sim, Gazebo, Unity, RoboDK*

CAD & Mechanisms: *Onshape, Fusion 360, linkage and manipulator design, kinematics, 3D printing*

Electronics & Prototyping: *KiCAD, motor controllers, stepper drivers, power conversion, microcontrollers (Arduino), soldering, bench testing, fusing and E-stop safety*